Nick DiSanto

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Research Interests

Computer Vision & Machine Learning: image restoration and reconstruction, inverse problems, 3D vision, neural representations, and geometric and probabilistic modeling

Education

Vanderbilt University Jan. 2025 – Present

Ph.D. in Computer Science, expected 2029

GPA: 3.96/4.0. Advised by Dr. Ipek Oguz. Research affiliated with VISE.

California Baptist University Aug. 2019 – Apr. 2023

B.S. in Computer Science, summa cum laude

GPA: 4.0/4.0. Ranked #1 in graduating class.

Research Experience

Vanderbilt Medical Image Computing Lab (MedICL) Jan. 2025 – Present

Graduate Student Researcher

- Developing deep learning methods for **3D medical image restoration** from incomplete or motion-corrupted observations, emphasizing preservation and recovery of anatomical structure under limited supervision
- Investigating **implicit neural representations (INRs)** for underconstrained 3D OCTA reconstruction, studying spectral bias and structural supervision for recovering missing vessel information
- Developing **probabilistic methods** for vascular connectivity restoration, using orientation-conditioned stochastic transport and continuous geometric fields to infer plausible connections between fragmented vessels
- Designed VAMOS-OCTA, combining vessel-weighted reconstruction with novel axial and lateral projection losses to preserve vascular detail and volumetric consistency in motion-corrupted OCTA
- Developed LaMa-depth-stack, adapting frozen 2D inpainting priors to 3D OCTA restoration by reorienting along en face slices, achieving the best reconstruction across two datasets by exploiting cross-slice redundancy

CalBaptist ML & NLP Lab Jan. 2022 – Dec. 2023

Lead Undergraduate Researcher

- Led **three research projects** end-to-end, from conception through implementation, analysis, and scientific writing
- Developed representation learning methods for large-scale geospatial social media data; in separate medical imaging work, designed contextual augmentation methods to improve generalization of classification models

Selected Publications

Reorienting 2D Inpainting for Robust Volumetric OCTA Restoration
N DiSanto, JJ Watson, YK Tao, I Oguz.
Accepted to SPIE Medical Imaging, 2027. **[Oral]** [code]

Multi-Scale Feature Adaptation Between a Frozen SAM 3 Encoder and a Convolutional Decoder in a U-Shaped Architecture for Medical Image Segmentation
EK Aghdam, **N DiSanto**, T Derr, JH Noble.
Accepted to SPIE Medical Imaging, 2027. **[Oral]** [code]

VAMOS-OCTA: Vessel-Aware Multi-Axis Orthogonal Supervision for Inpainting Motion-Corrupted OCT Angiography Volumes
N DiSanto, EK Aghdam, H Liu, JJ Watson, YK Tao, H Li, I Oguz.
SPIE Medical Imaging, 2026. **[Best Student Paper]** **[Oral Deep-Dive]** [pdf] [code]

VFMStitch: A Vision-Foundation-Model Empowered Framework for 3D Ultrasound Stitching via Geometric–Semantic Feature Fusion
X Yao, **N DiSanto**, R Yu, J Wang, D Lu, G Arenas, B Oguz, A Pouch, N Schwartz, BC Byram, I Oguz.
Medical Imaging with Deep Learning (MIDL), 2026. [pdf] [code]

Monocular Absolute Depth Estimation from Endoscopy via Domain-Invariant Feature Learning and Latent Consistency
H Li, D Lu, J d'Almeida, D Isik, EK Aghdam, **N DiSanto**, A Acar, S Sharma, JY Wu, RJ Webster III, I Oguz.
SPIE Medical Imaging, 2026. **[Oral]** [pdf] [code]

A Benchmark Study of Methods for Surgical Instrument Segmentation in Central Airway Endoscopy
D Isik, KJ Oguine, **N DiSanto**, J d'Almeida, RJ Webster III, I Oguz, H Li.
SPIE Medical Imaging, 2026. **[Best Poster: Honorable Mention]** [paper]

LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching
X Yao, R Yu, **N DiSanto**, EK Aghdam, KJ Oguine, D Lu, A Lou, J Wang, D Hu, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz.
Medical Imaging with Deep Learning (MIDL), 2025. **[Spotlight]** **[Oral]** [pdf] [code]

Industry Experience

GE Vernova Dec. 2023 – Jan. 2025
Software Engineer

- Led the technical roadmap and development of **Disruption Prepare's wildfire risk forecasting** from prototype to production, integrating satellite hotspots, wildfire reports, weather forecasts, and grid asset data into geospatial pipelines; designed hotspot-based proxy labeling to train asset-level risk models with limited ground truth
- Co-developed with Microsoft an **LLM-based risk synthesis layer**, combining model predictions and contextual signals into interpretable wildfire risk scores and using prior predictions to improve temporal stability
- Contributed to satellite-based vegetation risk analytics, evaluating spatial granularity for encroachment detection
- Co-led an **initiative integrating AI coding tools into workflows** and presented a live demo at a CEO-led company all-hands; it received full funding, and I was recognized with a **GE Vernova Changemaker Award**
- Mentored two undergraduate SWE interns, including sprint planning, code review, and technical guidance

GE Vernova Jun. – Aug. 2022; Jun. – Aug. 2023
Software Engineering Intern

- Built and deployed a **containerized data ingestion pipeline for industrial sensor data**, processing raw telemetry into team-specific outputs for downstream systems across different organizations
- Maintained and optimized **production data ingestion microservices** and containerized deployment pipelines

Keysight Technologies Jun. 2021 – Aug. 2021
Software Engineering Intern

- Built an internal team portal centralizing documentation, training materials, and shared resources
- Developed customer-facing **product configuration interfaces in Oracle CPQ/BML** and analyzed configuration logic to consolidate redundant options, reducing unnecessary configurations by **40%**

Technical Skills

Programming: Python, C++, Java, SQL, Bash
ML & GPU Computing: PyTorch, NumPy/SciPy, scikit-learn, OpenCV, CUDA, multi-GPU training/profiling
Systems & Infrastructure: Linux, SLURM, Git, Docker, Kubernetes, GitHub Actions, AWS/GCP/Azure

Teaching & Mentoring

Teaching Assistant / Tutor 2021 – 2023
Algorithms, Discrete Structures, Computer Architecture, Intro to C++, Physics for Engineers

Awards & Honors

SPIE Medical Imaging Best Student Paper Award Feb. 2026
GE Vernova Changemaker — selected from 1,200 nominees Nov. 2024
Computer Science Outstanding Student Award Apr. 2023
Alpha Chi Honor Society — inducted among top 10% of students Apr. 2022
Engineering Excellence Scholarship Jan. 2021 – Apr. 2023